ANTIPODAL CHORDS OF CLOSED CURVES AND THE EXPONENT THREE

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ABSTRACT. For a closed curve of length L, the chord joining two points at arclength distance $L/2$ is an antipodal chord. We prove that among all closed curves of length L in $\mathbb{R}^d$ the circle uniquely maximizes the p-th mean of the antipodal chord length for every $p \le 3$, and that the circle is not even a local maximizer when $p > 3$. This settles the half-length case of the mean-chord problem of Exner, Harrell and Loss, which was known only for $p \le 2$, and shows that the critical exponent there is 3 and not 5/2.

1. INTRODUCTION

Let $\Gamma\colon \mathbb{R} \to \mathbb{R}^d$ be a closed rectifiable curve of length L, parametrized by arclength, so that Γ is L-periodic and $|\Gamma'| = 1$ almost everywhere. For $0 < u < L$ and $p > 0$ consider the p-th mean of the chords that join points at arclength distance u,

$$c_p(\Gamma, u) = \frac{1}{L}\int_0^L |\Gamma(s+u) - \Gamma(s)|^p \, ds.$$

On the circle of length L every such chord has length $\frac{L}{\pi}\sin\frac{\pi u}{L}$. The question whether

$$c_p(\Gamma, u) \le \Big(\frac{L}{\pi}\sin\frac{\pi u}{L}\Big)^p \tag{1}$$

holds for all closed curves goes back to Lükő [8], who proved it for $p = 2$; Pech [9] gave another proof in all dimensions. Exner [4] and Exner, Harrell and Loss [6, 7] found that (1) governs two isoperimetric problems in mathematical physics: the shape of a loop of given length that maximizes the ground-state energy of a leaky quantum graph $-\Delta - \alpha\delta(x - \Gamma)$ in the plane, and the equilibrium shape of a uniformly charged loop; see also the review [5]. They proved (1) for every $p \le 2$ and asked for the largest exponent for which it remains true. By Jensen's inequality, validity for one exponent implies validity for every smaller one. A stadium shows that (1) fails for p above about 3.15 [6], and the numerical experiments of Abrams, Cantarella, Fu, Ghomi and Howard [1] point to a threshold between 3.3 and 3.58 for the double integral $\iint |\Gamma(s)-\Gamma(t)|^p \, ds\, dt$, which averages (1) over u. Exner, Fraas and Harrell [3] studied the second variation of $c_p(\cdot, u)$ at the circle and stated that the circle stops being a local maximizer when p exceeds $(4-\cos\frac{2\pi u}{L})/(1-\cos\frac{2\pi u}{L})$, a number equal to 5/2 for $u = L/2$. Beyond $p = 2$ no global result has been known for any u.

This paper settles the case $u = L/2$ completely. We call $\Gamma(s+L/2)-\Gamma(s)$ an antipodal chord.

Theorem A. *Let $d \ge 2$ and let Γ be a closed rectifiable curve of length L in $\mathbb{R}^d$. For every $0 < p \le 3$,*

$$\frac{1}{L}\int_0^L \Big|\Gamma\Big(s+\frac{L}{2}\Big) - \Gamma(s)\Big|^p \, ds \le \Big(\frac{L}{\pi}\Big)^p,$$

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with equality if and only if Γ *is a circle traversed once. For every* $p > 3$ *there are smooth closed plane curves of length* L*, arbitrarily close to the circle in* C^∞*, for which the inequality fails.*

The second part of Theorem A comes from an explicit deformation of the circle that works for every u.

Theorem B. *For* $0 < u < L$ *and* $p > 3/\sin^2(\pi u/L)$ *there are smooth closed plane curves of length* L*, arbitrarily close to the circle in* C^∞*, for which* (1) *fails. For* $u = L/2$ *the circle is the global maximizer for every* $p \le 3$*.*

Since $3/\sin^2(\pi u/L)$ is larger than $(4 - \cos\frac{2\pi u}{L})/(1 - \cos\frac{2\pi u}{L})$ for every u, Theorem B is consistent with the stability statement of [3] for p below their threshold. For $u = L/2$, however, Theorem A shows that the circle is the global maximizer for all p in $(5/2, 3]$, so the assertion of [3] that the circle fails to be a local maximizer for $p > 5/2$ is incorrect. The second variation along the deformation used below changes sign exactly at $p = 3/\sin^2(\pi u/L)$ (Proposition 7.1); this is the mode $n = 2$ of [3], whose coefficient there has the wrong sign between the two thresholds. Figure 1 compares the two curves.

The proof of Theorem A has three steps. First, half of the antipodal chord is a 1-Lipschitz loop W that satisfies $W(t + \pi) = -W(t)$ (Section 2). The problem becomes an optimal control problem: maximize $\int |W|^3$ over such loops. Second, the Pontryagin maximum principle turns a maximizer into an orbit of a central force problem, whose integrals of motion classify the candidates (Sections 3 and 4). Third, an extremal loop that is neither the circle nor too small to matter would need an apsidal angle equal to an odd multiple of π, and we show that the apsidal angle lies strictly between π and 3π (Section 5). The last step reduces to an explicit polynomial inequality, and it is there that the exponent 3 enters: for $p = 3$ the apsidal angle tends to π at both ends of the family of orbits.

2. Reduction to antipodal loops

By scaling we take $L = 2\pi$ in Sections 2–6. Write $|\cdot|$ for the Euclidean norm on $\mathbb{R}^d$ and

$$\mathcal{A} = \big\{w\colon [0,\pi] \to \mathbb{R}^d :\ |w(t) - w(t')| \le |t - t'| \text{ for all } t, t',\quad w(\pi) = -w(0)\big\},$$

$$J(w) = \int_0^\pi |w(t)|^3\, dt.$$

Every $w \in \mathcal{A}$ extends to a 2π-periodic 1-Lipschitz map $W\colon \mathbb{R} \to \mathbb{R}^d$ with $W(t + \pi) = -W(t)$; we use the same letter for a loop and its extension.

Lemma 2.1. *Let* Γ *be a closed curve of length* 2π *parametrized by arclength and put* $W(s) = \frac{1}{2}\big(\Gamma(s + \pi) - \Gamma(s)\big)$*. Then* $W|_{[0,\pi]} \in \mathcal{A}$ *and*

$$\frac{1}{2\pi}\int_0^{2\pi} |\Gamma(s + \pi) - \Gamma(s)|^3\, ds = \frac{8}{\pi}\, J(W).$$

If $|W'| = 1$ *almost everywhere, then* Γ *is a translate of* $-W$*.*

Proof. $W(s + \pi) = \frac{1}{2}(\Gamma(s + 2\pi) - \Gamma(s + \pi)) = -W(s)$, and $|W'| \le \frac{1}{2}(|\Gamma'(s + \pi)| + |\Gamma'(s)|) = 1$ almost everywhere. The integrand is $8|W|^3$, and $|W|$ is π-periodic. If $|W'(s)| = 1$, the two unit vectors $\Gamma'(s + \pi)$ and $\Gamma'(s)$ are at distance 2, hence opposite, and $W'(s) = -\Gamma'(s)$. □

Theorem A for $p = 3$ is therefore equivalent to $J(w) \le \pi$ on $\mathcal{A}$, with equality only for unit circles traversed at unit speed. The circle $w(t) = e_1 \cos t + e_2 \sin t$, with e_1, e_2 orthonormal, gives $J = \pi$.

Lemma 2.2. *If $w \in \mathcal{A}$, then $|w| \le \pi/2$. If moreover the extension W vanishes somewhere, then $J(w) \le \pi^4/32 < \pi$.*

Proof. For $t \in [0,\pi]$, $2|w(t)| \le |w(t) - w(0)| + |w(t) - w(\pi)| \le t + (\pi - t)$. If $W(t_0) = 0$, then $W(t_0 + \pi) = 0$, and $|W(t)| \le \min(t - t_0,\, t_0 + \pi - t)$ on $[t_0, t_0 + \pi]$. Since $|W|$ is π-periodic, $J(w) \le 2\int_0^{\pi/2} s^3\, ds = \pi^4/32$, and $\pi^3 < 32$. □

3. Existence and the maximum principle

Lemma 3.1. *J attains its maximum on $\mathcal{A}$, and the maximum is at least π.*

Proof. By Lemma 2.2, $\mathcal{A}$ is a uniformly bounded, equicontinuous family, closed under uniform convergence; by the Arzelà–Ascoli theorem it is compact in $C([0,\pi], \mathbb{R}^d)$, and J is continuous there. □

Let w be a maximizer. We view it as an optimal trajectory of the control system $\dot w = v$, $|v| \le 1$, with running payoff $f(w) = |w|^3$, which is of class C^2 with $\nabla f(w) = 3|w|w$, and endpoint constraint $w(\pi) + w(0) = 0$. The maximum principle for problems with constrained endpoints [10, Ch. I and II] gives a constant $p_0 \in \{0, 1\}$ and an absolutely continuous $p\colon [0,\pi] \to \mathbb{R}^d$, not both zero, such that almost everywhere

$$\dot p = -p_0\, \nabla f(w), \qquad \dot w \cdot p = \max_{|v| \le 1} v \cdot p = |p|, \tag{2}$$

the function $H = p_0 f(w) + |p|$ is constant, and the transversality condition holds. The tangent space of the endpoint constraint is $\{(h, -h)\}$, so transversality reads $p(0) \cdot h + p(\pi) \cdot h = 0$ for all h, that is, $p(\pi) = -p(0)$.

If $p_0 = 0$, then p is constant and $p(\pi) = -p(0)$ forces $p = 0$, which is excluded. Hence $p_0 = 1$. Extend p by $P(t + \pi) = -P(t)$ and write W for the extension of w. The system (2) is invariant under $(w, p) \mapsto (-w, -p)$ and the values at 0 and π match, so (W, P) is absolutely continuous and 2π-periodic on $\mathbb{R}$ and satisfies

$$\dot P = -3|W|W, \qquad \dot W = \frac{P}{|P|} \ \text{ where } P \neq 0, \qquad |W|^3 + |P| = E. \tag{3}$$

Lemma 3.2. *Either $J(w) \le \pi^4/32$, or $E > 0$, the bivector $\Lambda = W \wedge P$ is a nonzero constant, and W lies in the plane of Λ.*

Proof. If $E = 0$ then $W \equiv 0$. Let $E > 0$. The set where $P = 0$ is contained in $\{|W| = E^{1/3}\}$, on which $\dot P = -3|W|W \neq 0$; an absolutely continuous function has zero derivative almost everywhere on its zero set, so $\{P = 0\}$ is null. Almost everywhere $\frac{d}{dt}(W \wedge P) = \frac{P}{|P|} \wedge P - 3|W|\, W \wedge W = 0$, so Λ is constant.

Suppose $\Lambda = 0$. On the open set where $W \neq 0$ the unit vector $U = W/|W|$ satisfies $\dot U = (\dot W - (U \cdot \dot W)U)/|W| = 0$ almost everywhere, because $\dot W$ is parallel to P and hence to W. If W never vanished, U would be constant on $\mathbb{R}$, contradicting $U(t + \pi) = -U(t)$. So W vanishes somewhere and Lemma 2.2 applies.

If $\Lambda \neq 0$, then $W(t)$ and $P(t)$ span the plane of Λ for every t. □

4. The extremal orbits

Assume $\Lambda \neq 0$ and identify the plane of Λ with $\mathbb{C}$, oriented so that $\lambda = \operatorname{Im}(\overline{W} P) > 0$. Then $\lambda \le |W||P|$, so W and P never vanish, and (3) is a smooth system. Introduce the time τ with

$dt = |P|\,d\tau$.[1] Then

$$\frac{dW}{d\tau} = P, \qquad \frac{dP}{d\tau} = -3\big(E - |W|^3\big)|W|\,W,$$

which is Newton's equation in the central potential $V(\rho) = E\rho^3 - \frac{1}{2}\rho^6$, with energy $\frac{1}{2}|P|^2 + V(|W|) = \frac{1}{2}E^2$ and angular momentum λ. Write $W = \rho e^{i\psi}$ and

$$g(\rho) = \rho\,(E - \rho^3), \qquad 0 < \rho \le E^{1/3}.$$

The classical reduction of a central force problem [2, §8] gives $\rho^2\,d\psi/d\tau = \lambda$ and $\rho^2(d\rho/d\tau)^2 = g(\rho)^2 - \lambda^2$. Returning to t, and using $|P| = E - \rho^3$,

$$\Big(\frac{d\rho}{dt}\Big)^2 = 1 - \frac{\lambda^2}{g(\rho)^2}, \qquad \frac{d\psi}{dt} = \frac{\lambda}{\rho\,g(\rho)}. \tag{4}$$

The function g is strictly concave, with maximum $a_E = \frac{3}{4}E^{4/3}4^{-1/3}$ at $\rho_E = (E/4)^{1/3}$, and $g(\rho) \ge \lambda$ along the orbit. Two cases occur.

Circular orbits. If $\lambda = a_E$, then $\rho \equiv \rho_E$, $E = 4\rho_E^3$ and $d\psi/dt = 1/\rho_E$, so W runs around the circle of radius ρ_E at unit speed.

Oscillating orbits. If $0 < \lambda < a_E$, the equation $g(\rho) = \lambda$ has two simple roots $\rho_1 < \rho_E < \rho_2$, and ρ oscillates periodically between them. Let T be the period of ρ in the time t and Ψ the increase of ψ over one period, the apsidal angle. By (4) and the symmetry of the orbit about its apsidal lines,

$$T = 2\int_{\rho_1}^{\rho_2} \frac{g(\rho)\,d\rho}{\sqrt{g(\rho)^2 - \lambda^2}}, \qquad \Psi = 2\int_{\rho_1}^{\rho_2} \frac{\lambda\,d\rho}{\rho\sqrt{g(\rho)^2 - \lambda^2}}. \tag{5}$$

The substitution $\rho = E^{1/3}r$, $\lambda = E^{4/3}k$ shows that Ψ depends only on $k = \lambda E^{-4/3} \in (0, a)$, where $a = a_1 = \frac{3}{4}\,4^{-1/3}$, and that $T = E^{1/3}T_1(k)$, where T_1 is the period for $E = 1$.

5. The apsidal angle

In this section $E = 1$, $g(\rho) = \rho - \rho^4$ on $(0, 1)$, and $0 < k < a$.

Lemma 5.1. *$\pi < \Psi(k) < 3\pi$ for every $k \in (0, a)$.*

The proof rests on writing Ψ against an explicit kernel whose apsidal angle is exactly π.

Step 1: the level variable. Let $\rho_-(y) < \rho_+(y)$ be the two roots of $g(\rho) = y$ for $0 < y < a$. Splitting (5) at ρ_E and substituting $y = g(\rho)$ on each branch, with $d\rho/dy = 1/g'(\rho) = 1/(1-4\rho^3)$,

$$\Psi(k) = 2\int_k^a \frac{k\,D(y)}{\sqrt{y^2 - k^2}}\,dy, \qquad D(y) = \frac{1}{\rho_-(1 - 4\rho_-^3)} + \frac{1}{\rho_+(4\rho_+^3 - 1)} > 0. \tag{6}$$

Step 2: the comparison kernel. Put $D_0(y) = a/\big(y\sqrt{a^2 - y^2}\big)$. The substitution $z = y^{-2}$ turns $\int_k^a dy/\big(y\sqrt{(a^2 - y^2)(y^2 - k^2)}\big)$ into $\frac{1}{2}\int_{a^{-2}}^{k^{-2}} dz/\sqrt{(a^2z - 1)(1 - k^2z)} = \pi/(2ak)$, so

$$2\int_k^a \frac{k\,D_0(y)}{\sqrt{y^2 - k^2}}\,dy = \pi, \qquad \text{hence} \qquad \Psi(k) - \pi = 2k\int_k^a \frac{D(y) - D_0(y)}{\sqrt{y^2 - k^2}}\,dy.$$

It remains to compare $X = yD(y)$ with $Y = yD_0(y) = a/\sqrt{a^2 - y^2}$.[2]

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[1] By Lemma 2.2, $|P| \ge \lambda/|W| \ge 2\lambda/\pi$, so τ is a smooth increasing function of t.

[2] The kernel D_0 is the solution of the Abel equation $2\int_k^a kD_0(y)(y^2 - k^2)^{-1/2}dy = \pi$ for all k; it describes a medium in which every orbit has apsidal angle π.

Step 3: a rational parametrization. Write $\rho_\mp = \sigma \mp \delta$. Subtracting $\rho_- - \rho_-^4 = \rho_+ - \rho_+^4$ gives $(\rho_- + \rho_+)(\rho_-^2 + \rho_+^2) = 1$, that is

$$\delta^2 = \frac{1 - 4\sigma^3}{4\sigma}, \qquad \tfrac{1}{2} < \sigma < 4^{-1/3},$$

and a short computation gives

$$\rho_-\rho_+ = \frac{8\sigma^3 - 1}{4\sigma}, \qquad (1 - 4\rho_-^3)(4\rho_+^3 - 1) = \frac{(1 - 4\sigma^3)(128\sigma^6 + 1)}{4\sigma^3}, \qquad y = \frac{64\sigma^6 - 1}{16\sigma^2}.$$

The map $\sigma \mapsto y$ is an increasing bijection of $(\frac{1}{2}, 4^{-1/3})$ onto $(0, a)$. Since $\rho_+ - \rho_+^4 = \rho_- - \rho_-^4$, we have $\rho_+(1 - 4\rho_+^3) - \rho_-(1 - 4\rho_-^3) = -3(\rho_+ - \rho_-) = -6\delta$; bringing the two terms of D to a common denominator gives

$$D = \frac{6\delta}{\rho_-\rho_+(1 - 4\rho_-^3)(4\rho_+^3 - 1)} = \frac{96\,\sigma^4\delta}{(8\sigma^3 - 1)(1 - 4\sigma^3)(128\sigma^6 + 1)}.$$

Now set $\xi = 4^{1/3}\sigma \in (2^{-1/3}, 1)$. Then $y/a = (4\xi^6 - 1)/(3\xi^2)$, and

$$X^2 = \frac{9\xi^3(2\xi^3 + 1)^2}{4(1 - \xi^3)(8\xi^6 + 1)^2}, \tag{7}$$

$$Y^2 = \frac{9\xi^4}{9\xi^4 - (4\xi^6 - 1)^2} = \frac{9\xi^4}{(1 - \xi^2)(2\xi^2 + 1)^2(4\xi^6 + 3\xi^2 - 1)}. \tag{8}$$

Subtracting (8) from (7) and factoring,

$$X^2 - Y^2 = \frac{9\xi^3(2\xi^3 - 1)\,M(\xi)}{4(1 + \xi)(2\xi^2 + 1)^2(1 + \xi + \xi^2)(4\xi^4 - 2\xi^2 + 1)^2(4\xi^6 + 3\xi^2 - 1)}, \tag{9}$$

where

$$M(\xi) = 1 + 6\xi + 7\xi^2 - 2\xi^3 - 4\xi^4 + 18\xi^5 + 36\xi^6 + 16\xi^7 - 16\xi^8 - 8\xi^9.$$

Proof of Lemma 5.1. For $0 < \xi \le 1$ we have $\xi^3 \le \xi$, $\xi^4 \le \xi^2$, $\xi^8 \le \xi^6$ and $\xi^9 \le \xi^7$, so

$$M(\xi) \ge 1 + 4\xi + 3\xi^2 + 18\xi^5 + 20\xi^6 + 8\xi^7 > 0, \qquad M(\xi) \le 1 + 6 + 7 + 18 + 36 + 16 = 84.$$

On $(2^{-1/3}, 1)$ the factor $2\xi^3 - 1$ is positive, and so is every factor of the denominator of (9); note $4\xi^4 - 2\xi^2 + 1 = (2\xi^2 - \frac{1}{2})^2 + \frac{3}{4}$. Hence $X > Y$, so $D > D_0$ on $(0, a)$ and $\Psi(k) > \pi$ by Step 2.

For the upper bound, each factor of the denominator of (9) increases on $[2^{-1/3}, 1]$,[3] and at $\xi = 2^{-1/3}$, where $\xi^2 > 0.62$ and $\xi^6 = \frac{1}{4}$, the denominator exceeds $4 \cdot 1.79 \cdot 5.01 \cdot 2.41 \cdot 1.68 \cdot 1.86 > 252$. The numerator is at most $9 \cdot 84 = 756$. Thus $X^2 - Y^2 < 3$. Since $Y \ge 1$,

$$0 < X - Y = \frac{X^2 - Y^2}{X + Y} < \frac{3}{2},$$

and therefore

$$\Psi(k) - \pi = 2k \int_k^a \frac{X - Y}{y\sqrt{y^2 - k^2}}\,dy < 3k \int_k^a \frac{dy}{y\sqrt{y^2 - k^2}} = 3\,\operatorname{arcsec}\frac{a}{k} < \frac{3\pi}{2}. \qquad \square$$

Remark 5.2. Lemma 5.1 is where $p = 3$ is special. For the payoff $|w|^p$ the same reduction gives $g(\rho) = \rho(E - \rho^p)$, and the apsidal angle of nearly circular orbits tends to $2\pi/\sqrt{p+1}$, which equals π exactly when $p = 3$. For $p > 3$ this limit is smaller than π, and the instability of Section 7 sets in.

[3] The derivative of $4\xi^4 - 2\xi^2 + 1$ is $4\xi(4\xi^2 - 1) > 0$ for $\xi > \frac{1}{2}$.

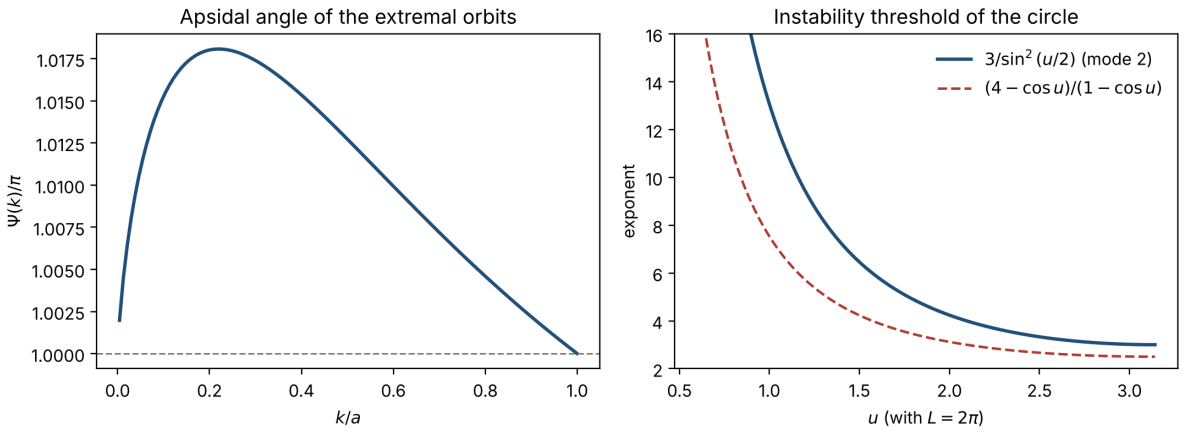


FIGURE 1. Left: the apsidal angle $\Psi(k)$ of the extremal orbits, computed from (5); it tends to π at both ends and its maximum is about 1.018π. Right: the exponent $3/\sin^2(u/2)$ above which the circle is unstable (Proposition 7.1) and the exponent $(4-\cos u)/(1-\cos u)$ of [3], for $L = 2\pi$.

6. Proof of Theorem A

Let w be a maximizer of J on $\mathcal{A}$ (Lemma 3.1), and assume $J(w) > \pi^4/32$. By Lemma 3.2, its extension W is a planar orbit of Section 4 with $\lambda > 0$, and $W(t+\pi) = -W(t)$. Since $|P| = E - \rho^3 > 0$, we have $\rho \le E^{1/3}$ and hence $J(w) \le \pi E$.

Circular orbits. Over a time interval of length π the angle increases by π/ρ_E, and $W(t+\pi) = -W(t)$ requires $\pi/\rho_E \in \pi + 2\pi\mathbb{Z}$, so $\rho_E = 1/(2j+1)$ with $j \ge 0$. Then $J(w) = \pi\rho_E^3$, which equals π for $j = 0$ and is smaller otherwise.

Oscillating orbits. The function ρ is π-periodic with least period T, so $\pi = mT$ for an integer $m \ge 1$. Starting at an apocentre, after time π the point W is again at an apocentre, rotated by $m\Psi$; since it equals $-W$ there, $m\Psi \in \pi + 2\pi\mathbb{Z}$. For $m = 1$ this contradicts Lemma 5.1. Let $m \ge 2$. In the normalization $E = 1$, let Q be a pericentre and A the next apocentre. They lie at radii ρ_1, ρ_2 and subtend the angle $\Psi/2 \in (\pi/2, 3\pi/2)$ at the origin, so $\cos(\Psi/2) < 0$ and

$$\frac{T_1}{2} \ge |A - Q| \ge \sqrt{\rho_1^2 + \rho_2^2} = \frac{1}{\sqrt{\rho_1 + \rho_2}} \ge 2^{-1/6},$$

using $(\rho_1+\rho_2)(\rho_1^2+\rho_2^2) = 1$ (Step 3) and $(\rho_1+\rho_2)^3 \le 2(\rho_1+\rho_2)(\rho_1^2+\rho_2^2) = 2$. Hence $T_1 \ge 2^{5/6}$, and $\pi = mE^{1/3}T_1 \ge 2^{11/6}E^{1/3}$, so

$$J(w) \le \pi E \le \pi\Big(\frac{\pi}{2^{11/6}}\Big)^3 < 0.69\,\pi.$$

In every case other than the unit circle, $J(w) < \pi$. Since the maximum is at least π, it equals π, and every $w \in \mathcal{A}$ with $J(w) = \pi$ is a maximizer, hence a unit circle traversed at unit speed.

By Lemma 2.1 and scaling, this proves Theorem A for $p = 3$, including the equality case: equality forces $|W'| = 1$, so Γ is a translate of $-W$, a circle traversed once. For $0 < p < 3$ Jensen's inequality gives

$$\Big(\frac{1}{L}\int_0^L |\Gamma(s+L/2) - \Gamma(s)|^p\,ds\Big)^{1/p} \le \Big(\frac{1}{L}\int_0^L |\Gamma(s+L/2) - \Gamma(s)|^3\,ds\Big)^{1/3} \le \frac{L}{\pi},$$

and equality throughout forces equality for $p = 3$. The statement for $p > 3$ is the case $u = L/2$ of Proposition 7.1. □

7. Instability of the circle

Proposition 7.1. *Let $L = 2\pi$, $0 < u < 2\pi$, $S = \sin^2(u/2)$, and let Γ_ε be the closed curve with $\Gamma_\varepsilon'(s) = e^{i(s+\varepsilon\sin 2s)}$. Then Γ_ε has length 2π and, as $\varepsilon \to 0$,*

$$\frac{1}{2\pi}\int_0^{2\pi} |\Gamma_\varepsilon(s+u) - \Gamma_\varepsilon(s)|^p \, ds = (2\sin\tfrac{u}{2})^p\Big(1 + \frac{pS(pS-3)}{9}\,\varepsilon^2 + O(\varepsilon^3)\Big).$$

In particular (1) *fails for small $\varepsilon \neq 0$ whenever $p > 3/S$.*

Proof. With the Bessel expansion $e^{i\varepsilon\sin 2\tau} = \sum_{k\in\mathbb{Z}} J_k(\varepsilon)e^{2ik\tau}$, the curve closes because only odd frequencies occur in Γ_ε', and

$$\Gamma_\varepsilon(s+u) - \Gamma_\varepsilon(s) = \sum_k J_k(\varepsilon)e^{i(2k+1)s}\,\frac{e^{i(2k+1)u}-1}{i(2k+1)} = 2\sin\tfrac{u}{2}\,e^{i(s+u/2)}\,F(w),$$

where $w = e^{i(2s+u)}$, $\beta_k = \sin\frac{(2k+1)u}{2}\big/\big((2k+1)\sin\frac{u}{2}\big)$ and $F(w) = \sum_k J_k(\varepsilon)\beta_k w^k$. Using $J_0 = 1 - \frac{\varepsilon^2}{4} + O(\varepsilon^4)$, $J_{\pm1} = \pm\frac{\varepsilon}{2} + O(\varepsilon^3)$, $J_{\pm2} = \frac{\varepsilon^2}{8} + O(\varepsilon^4)$, $\beta_{-1} = 1$, $\beta_{-2} = \beta_1$ and the rapid decay of J_k,

$$F = 1 + \frac{\varepsilon}{2}(\beta_1 w - \bar w) + \varepsilon^2\Big(-\frac14 + \frac{\beta_2 w^2 + \beta_1 \bar w^2}{8}\Big) + O(\varepsilon^3)$$

uniformly in w. Writing $w = e^{i\theta}$, $|F|^2 = 1 + X$ with

$$X = \varepsilon(\beta_1 - 1)\cos\theta + \varepsilon^2\Big(\frac{\beta_1^2 - 1}{4} + \frac{\beta_2 - \beta_1}{4}\cos 2\theta\Big) + O(\varepsilon^3).$$

Averaging over θ, $\overline{X} = \frac{\beta_1^2-1}{4}\varepsilon^2 + O(\varepsilon^3)$ and $\overline{X^2} = \frac{(\beta_1-1)^2}{2}\varepsilon^2 + O(\varepsilon^3)$, so

$$\overline{|F|^p} = 1 + \frac{p}{2}\overline{X} + \frac{p}{4}\Big(\frac{p}{2} - 1\Big)\overline{X^2} + O(\varepsilon^3) = 1 + \frac{p(1-\beta_1)}{16}\big((p-2)(1-\beta_1) - 2(1+\beta_1)\big)\varepsilon^2 + O(\varepsilon^3).$$

Finally $\beta_1 = \sin\frac{3u}{2}/(3\sin\frac{u}{2}) = 1 - \frac43 S$, which gives the coefficient $pS(pS-3)/9$. □

For $u = \pi$ the coefficient is $p(p-3)/9$; it vanishes at $p = 3$, where the circle is a degenerate but global maximizer by Theorem A. For $p = 2$ and every u it equals $-\frac{2S(3-2S)}{9} < 0$, as Lükő's inequality requires.

Declarations

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Data availability. No data were generated. The figure is computed from the integrals (5). Scripts checking the algebra of Section 5 and the numerics are archived at `https://doi.org/10.5281/zenodo.24128091`.
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